

Saturation of the Emergent Electric Field Makes the Photon Anharmonic at Lattice Momenta

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The emergent photon of quantum spin ice is described at long wavelength by Gaussian lattice electrodynamics, a harmonic theory whose two parameters enter the transverse spectrum only through a common prefactor. That rigidity fixes three statements at every momentum and for every choice of parameters: the single-mode state $S^z(\mathbf{q})|0\rangle$ is exactly an eigenstate, so Feynman’s ratio saturates at $f_1/S = \omega$; each transverse channel carries a single spectral pole; and the two transverse polarizations are degenerate, which we verify by diagonalizing the lattice curl at 153 allowed momenta of three tori. Compactness must break all three, because the emergent electric field is saturated at $S^z = \pm\frac{1}{2}$: a field wave cannot be built by raising an amplitude, only by rearranging which links carry which of the two allowed values, and exponentially many ice configurations carry the same wave. We compute the operator-resolved $S^{zz}(\mathbf{q}, \omega)$ on 48-, 64-, 72- and 96-site clusters, the largest a constrained sector of dimension 1.47×10^8 , and find all three identities violated by order one and all three most strongly at the zone-boundary L point, where the variational bound overshoots by 1.68, a third of the channel weight leaves the dominant pole, and the doublet splits by 1.69. The bound overshoot ranks the five resolved momenta exactly as the softening does. The mechanism is exact and parameter-free: in the ice basis the probe is diagonal, so a ring exchange either shifts the density-wave amplitude by $a(c') - a(c) = \pm g_p$ or leaves it untouched, and the hexagons that leave it untouched are those lying in the wavefronts—one family in four when $\mathbf{q}$ is parallel to a $\langle 111 \rangle$ axis. Flipping those costs no spectral weight, and the soft eigenstate exploits it, recovering 89% of the dephasing the rigid wave suffers on the driven plaquettes and paying $0.108K$ there against its partner’s $1.082K$. A counterfactual that removes the cancellation makes L an ordinary momentum; at X , where no family is dark, nothing softens. The result is not a new excitation but the photon itself reorganized, and it predicts a two-feature zone-boundary response whose 0.55–0.59 energy ratio and two-to-one weight ratio are independent of the overall energy scale.

Introduction.—Quantum spin ice realizes a compact $U(1)$ lattice gauge theory in a magnetic insulator. In the two-in-two-out manifold of the pyrochlore lattice, virtual exchange resonates the six spins around an elementary hexagon, and the deconfined phase supports a gapless emergent photon together with gapped electric and magnetic charges [1–6]. At long wavelength this photon is described by Gaussian lattice electrodynamics, a theory of decoupled harmonic oscillators, and that description is quantitatively established and underpins the search for candidate materials [7–16].

Being harmonic makes the Gaussian theory unusually rigid, and that rigidity is what we exploit. It has two free parameters, an electric stiffness U and the ring coupling K , and they enter the transverse spectrum only through a common prefactor, $\omega_\lambda(\mathbf{q}) = \sqrt{2UK} \sigma_\lambda(\mathbf{q})$. Consequently three statements hold at every momentum for *every* choice of those parameters: the single-mode state $S^z(\mathbf{q})|0\rangle$ is exactly an eigenstate, so Feynman’s ratio saturates, $f_1/S = \omega$; each transverse channel has a single spectral pole; and the two transverse polarizations are degenerate. No renormalization of a harmonic theory can violate any of them. They are therefore a parameter-free baseline against which departures can be measured without a fit.

Compactness must violate all three, and for a reason that is structural rather than perturbative. The emer-

gent electric field is microscopically saturated, $E_\ell = S_\ell^z = \pm\frac{1}{2}$. A field wave at wavevector $\mathbf{q}$ therefore cannot be built by increasing an amplitude—there is no amplitude to increase—but only by rearranging which links carry which of the two allowed values. That rearrangement is not unique: exponentially many ice configurations share the same density-wave amplitude, so the excitation retains internal freedom at fixed spectral weight. A harmonic theory has no such freedom, which is precisely why its single-mode state is already exact. What is not obvious, and what we compute, is whether the excitation uses that freedom, by how much, and where.

We find that it does, dramatically, at zone-boundary momenta. Our results are:

1. All three harmonic identities fail by order one, and all three fail hardest at the zone-boundary L point: the Feynman bound overshoots the true level by 1.68, a third of the channel weight leaves the dominant pole, and the transverse doublet splits by 1.69.
2. The departure from harmonicity is not scattered. Ranked across the five momenta we resolve, the bound overshoot orders them *exactly* as the softening does, without inversion.
3. The mechanism is exact and requires no fitted parameter. In the ice basis the probe is diagonal, so

a ring exchange either shifts the density-wave amplitude or does not; the hexagons that do not are those lying in the wavefronts, one family in four when $\mathbf{q}$ is parallel to a $\langle 111 \rangle$ axis. Flipping them is motion at constant probe value, hence free of spectral-weight cost.

4. The soft eigenstate exploits exactly this. It recovers 89% of the dephasing that the rigid wave suffers on the driven plaquettes, paying only $0.108K$ there against its partner's $1.082K$ —a factor of ten, against an overall energy ratio of 0.59.
5. The effect is momentum-specific and size-stable: removing the geometric cancellation in a counterfactual makes L an ordinary momentum, no family is dark at X and nothing softens there, and the zone-boundary branch turns over on all three clusters we can diagonalize.

Model, observable, method.—We study the standard ring Hamiltonian on the spin-ice manifold [1, 8], $H = -K \sum_p (W_p + W_p^\dagger)$ with $W_p = S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^-$ over contractible elementary hexagons; in gauge language $W_p = e^{iB_p}$ and H is the compact magnetic energy $-2K \sum_p \cos B_p$. It has no diagonal part, a fact we use repeatedly. Since S^z is the emergent electric field the natural probe is $S^{zz}(\mathbf{q}, \omega) = N^{-1} \sum_{\mu, n} |\langle n | A_{\mathbf{q}}^\mu | 0 \rangle|^2 \delta(\omega - \omega_n)$, with $A_{\mathbf{q}}^\mu = \sum_{i \in \mu} e^{-i\mathbf{q} \cdot \mathbf{r}_i} S_i^z$ resolved by sublattice μ . All spectral fractions are normalized within S^{zz} at fixed $\mathbf{q}$. This is the theoretical pseudospin response, not a material-specific neutron cross section; the Supplemental Material treats the difference in full.

We evaluate S^{zz} by complete diagonalization at 48 sites and by Lanczos continued fractions at 64, 72 and 96, the last in a winding sector of dimension 146 758 352. Sectors are certified exhaustively at 48 and 72 sites and by the twelve largest of 519 at 96; Krylov dimension 200 against 120 leaves the dominant poles unchanged to machine precision. Details and convergence are in the Supplemental Material.

The harmonic baseline is rigid.—The gauge field lives on the links of the diamond lattice, equivalently on the pyrochlore sites, four per cubic cell, while the gradient that generates gauge transformations has one column per diamond site, two per cell. Gauge invariance $Cd_0 = 0$ —which we verify to 0.0×10^0 —therefore removes two of the four link modes and leaves exactly two transverse polarizations, the same count as Maxwell in three dimensions and for the same reason.

Their degeneracy is a separate fact and does not follow from that counting. We verified it directly by diagonalizing the lattice curl at every allowed momentum of fcc(4, 4, 4), fcc(3, 4, 5) and fcc(2, 3, 5): at all 153 momenta, including 32 of generic symmetry on the anisotropic boxes, $\sigma_2/\sigma_1 = 1.00000000$, with $\sigma_3/\sigma_1 \sim 10^{-15}$ confirming rank two. Because U and K multiply

both members alike, no renormalization of the Gaussian theory can split the doublet at all. Together with the two harmonic identities noted above, this fixes the baseline $(H_1, H_2, H_3) = (1, 0, 1)$ of Table I at every momentum and for every parameter choice.

Measuring the failure.—Table I collects the three departures at every momentum we resolve at 48 sites. All are order one, all are largest at L , and the first—the overshoot of the variational bound, which is precisely how far the single-mode state is from being an eigenstate—orders the five momenta exactly as the softening does.

Two of these deserve emphasis. That $H_2 = 0.335$ means a third of the weight in that channel has left its dominant pole: the response is not a single renormalized quasiparticle even qualitatively. And $H_3 = 1.691$ is a splitting of a doublet that the Gaussian theory holds exactly degenerate for any parameters, so it is not a large correction within that description but a failure of it. That interactions split these polarizations at intermediate wavevector was predicted semiclassically by Kwasiogoch, Douçot and Castelnovo [17]; what is new here is its size at the zone boundary, that only one member softens, and the microscopic reason.

Why compactness permits this.—The mechanism follows from one observation. In the basis of ice configurations $|c\rangle$ the probe is diagonal, $A_{\mathbf{q}}^\mu |c\rangle = a^\mu(c) |c\rangle$, whereas the ring exchange is purely off-diagonal, taking c to c' with hexagon p flipped. The commutator $[A_{\mathbf{q}}^\mu, W_p] = g_p^\mu(\mathbf{q}) W_p$ is therefore a statement about configurations,

$$a^\mu(c') - a^\mu(c) = \pm g_p^\mu(\mathbf{q}), \quad g_p^\mu = \sum_{j \in p \cap \mu} (-1)^j e^{-i\mathbf{q} \cdot \mathbf{r}_j}, \quad (1)$$

so g_p^μ is exactly the amount by which flipping p shifts the density-wave amplitude. We verified Eq. (1) on 4000 configurations, every hexagon and every sublattice: the largest discrepancy is 0.0×10^0 .

This turns a dynamical question into a geometric one. A pyrochlore hexagon is planar with its plane perpendicular to one of the four $\langle 111 \rangle$ axes, and each hexagon omits

TABLE I. Departures from harmonic lattice electrodynamics, 48 sites, evaluated in the softer transverse channel. A harmonic theory gives $H_1 = 1$, $H_2 = 0$, $H_3 = 1$ at every momentum for any U and K . H_1 is monotone in the softening; H_2 and H_3 are largest at L but not monotone.

$\mathbf{q}$	ω_{low}/K	H_1 $\frac{f_1/S}{\omega}$	H_2 off-pole	H_3 $\frac{\omega_{\text{hi}}}{\omega_{\text{lo}}}$	$\omega_{\text{low}}/\omega_G$
L	1.024	1.678	0.335	1.691	0.503
$(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$	1.744	1.346	0.216	1.187	0.824
X	2.235	1.274	0.255	1.531	0.952
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	1.895	1.260	0.164	1.045	1.076
$(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$	2.533	1.213	0.270	1.061	1.114

one sublattice, so the hexagons fall into four families. A plane wave is constant on planes perpendicular to $\mathbf{q}$; if a family's plane *is* such a wavefront, all six of its sites share a phase, the alternating sum cancels term by term, and $g_p^\mu = 0$ for every μ . At L , where $\mathbf{q} \parallel [1\bar{1}1]$, exactly one family satisfies this. Its measured geometric weight is 0.0000 against 96.0000 for each of the other three, and its axis is parallel to $\mathbf{q}$ to six digits, $|\hat{\mathbf{n}} \cdot \hat{\mathbf{q}}| = 1.000000$. We call it *dark*.

Now the consequence. Because H has no diagonal part, energy is ring coherence and nothing else, and every excitation carries an exact additive bill, $\omega_n = K \sum_p [\langle 0 | W_p + W_p^\dagger | 0 \rangle - \langle n | W_p + W_p^\dagger | n \rangle]$. Writing the excitation as the ground state modulated, $\psi(c) \simeq a(c)\psi_0(c)$, its coherence on p is a sum over the connected pairs of $a^*(c')a(c)\psi_0(c')\psi_0(c)$. Since $\psi_0 > 0$, the ground-state version of that sum interferes constructively—that is what makes the ground state cheap. In the excited state, by Eq. (1), a dark plaquette has $a(c') = a(c)$ and the factor is $|a|^2 > 0$: the pair still resonates in phase and *no* coherence is lost. A bright plaquette has $a(c') \neq a(c)$, the pair dephases, and the lost coherence is the energy. This is also what the f -sum rule has been saying: $f_1 = \frac{K}{2N} \sum_p \langle W_p + W_p^\dagger \rangle \sum_\mu |g_p^\mu|^2$ is a sum of dephasings, and it is blind to the dark family because a dark plaquette cannot be dephased.

The state uses the freedom.—Availability is a property of the momentum; the dark family is equally available to every level at L . What distinguishes the eigenstates is what they do with it, and the exact ledger settles that (Fig. 1a). The rigid single-mode state costs $1.846K$, of which $0.968K$ is dephasing on the bright families. The soft eigenstate costs $1.024K$, of which only $0.108K$ is bright: it has recovered 89% of the dephasing while leaving its dark-family cost essentially unchanged, so the entire relaxation of $0.822K$ is bright-family recovery (105% of it, the dark contribution being -5%). Its partner at the same momentum recovers nothing and pays $1.082K$ there, a factor of ten more, against an overall energy ratio of 0.59. The same pattern holds at 72 sites, where the bright-family ratio is 0.282 against an overall 0.589.

The soft mode is therefore a density wave dressed so as to re-phase the resonances the wave would otherwise dephase, depositing what remains on the family where flipping does not change the probe value and so costs no spectral weight. It buys energy in a currency the probe cannot see—and saturation of E is what makes that currency exist.

Controls.—Three checks establish that this is the operative mechanism rather than a description of it. At X no $\langle 111 \rangle$ axis is parallel to $\mathbf{q}$, the geometric weights are (96, 96, 96, 96), no family is dark: there is nowhere to deposit the disturbance and nothing softens, the lowest level sitting at 0.952 of its Gaussian energy against 0.503 at L . In a counterfactual in which the dark family is given the mean bright weight, $f_1(L)$ rises from 0.6835 to

0.8797—which coincides with $f_1(X) = 0.8797$, confirming that the entire suppression at L is the dark family—and L 's predictor becomes 1.168, tied with $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$ at 1.177, a momentum that softens only to 0.824. Remove the cancellation and L stops being special. Finally the ground state anticipates all of it: $S(L) = 0.3704$ is the largest structure factor of any resolved momentum, and f_1/S normalized by the Gaussian energy ranks the five momenta exactly as the measured softening does, a prediction available before any excited state is computed and carried to 2048 spins by sign-free Green's-function Monte Carlo, where the L -centered minimum persists with 1.7% drift (Fig. 4).

Consequences and robustness.—The mechanism bends the dispersion. Walking outward along $\Gamma \rightarrow L$ the branch first rises like a photon—at $t = \frac{1}{3}$ it lies *above* the Gaussian curve and is nearly degenerate with its partner, 1.895 against 1.980—then turns over and falls into the zone boundary, by 46%, 42% and 31% at 48, 64 and 72 sites, against the partner's 12.6% (Fig. 2). Each cluster resolves one interior momentum on that line, so what is established is the inequality $\omega(\text{interior}) > \omega(L)$ at three independent sizes rather than the shape of a minimum.

The scale-free measures are the stable ones (Table II, Fig. 3): the branch ratio is 0.55–0.59 across a factor 1.6×10^4 in sector dimension and three cluster shapes, and the soft branch holds 50–54% of the response throughout. Absolute energies drift by about ten percent, as expected at these sizes. Not everything is stable and we say which: the polarization splitting at X falls from 42% at 48 sites to 6% at 64, and we draw no conclusion from it.

Relation to the roton.—The phenomenology is Feynman's, and it is worth saying exactly how far the parallel runs. Our f_1/S is literally his construction; $S(\mathbf{q})$ peaks where the mode softens, as it does at the helium roton; his bound overshoots by 1.80 here against roughly a factor two in helium, and in both cases relaxation below it is the physics. The dispersion turns over. And, as with the helium roton, this is not a new quasiparticle: it is one of the two transverse polarizations, continuously connected to the photon doublet at small $\mathbf{q}$, carrying no new quantum number—phonon and roton are likewise one branch.

TABLE II. Size dependence at L . The scale-free measures—the branch ratio and the weight fraction—are the stable ones.

sites	48	64	72	96
sector dimension	9104	2.5×10^5	9.7×10^5	1.5×10^8
ω_{low}/K	1.024	0.954	1.068	0.904
weight	50.8%	50.0%	54.0%	54.4%
ω_{high}/K	1.730	1.652	1.814	1.647
weight	32.3%	31.3%	17.8%	22.5%
$\omega_{\text{low}}/\omega_{\text{high}}$	0.592	0.578	0.589	0.549
turnover to L	−46%	−42%	−31%	—

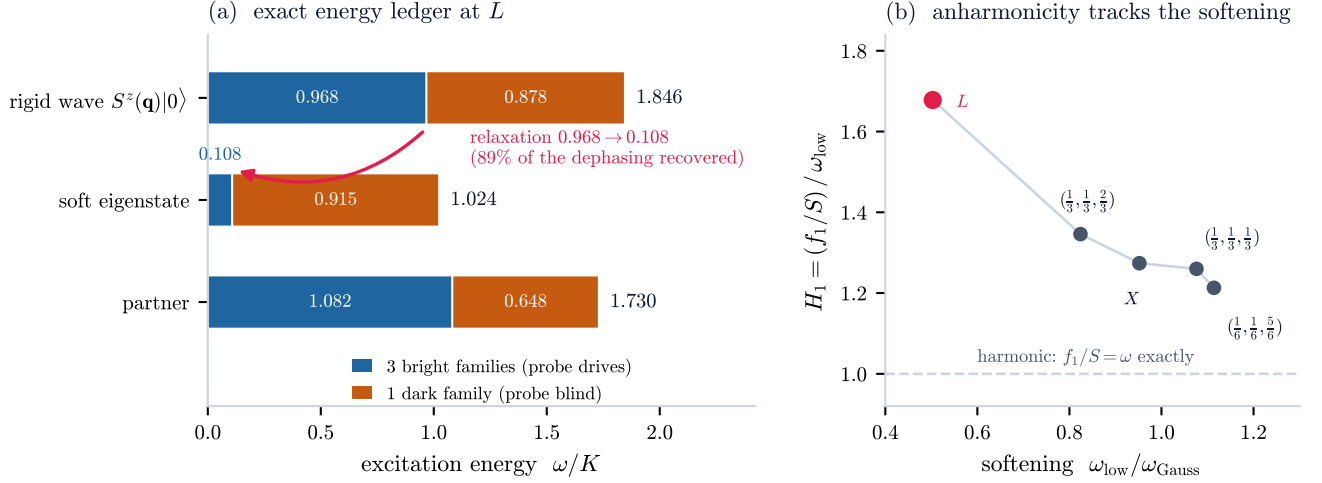


FIG. 1. **The mechanism and the anharmonicity it produces.** (a) Exact energy ledger at L , 48 sites, split by whether the probe drives the family. The rigid single-mode state pays 0.968 K of dephasing on the three bright families; the true soft eigenstate has recovered 89% of it while its dark-family cost is essentially unchanged, so the entire relaxation happens on the driven plaquettes. Its partner performs no such recovery and pays ten times more there. (b) In a harmonic theory $H_1 = (f_1/S)/\omega$ equals unity at every momentum for any parameters. Measured, it ranks the five resolved momenta exactly as the softening does.

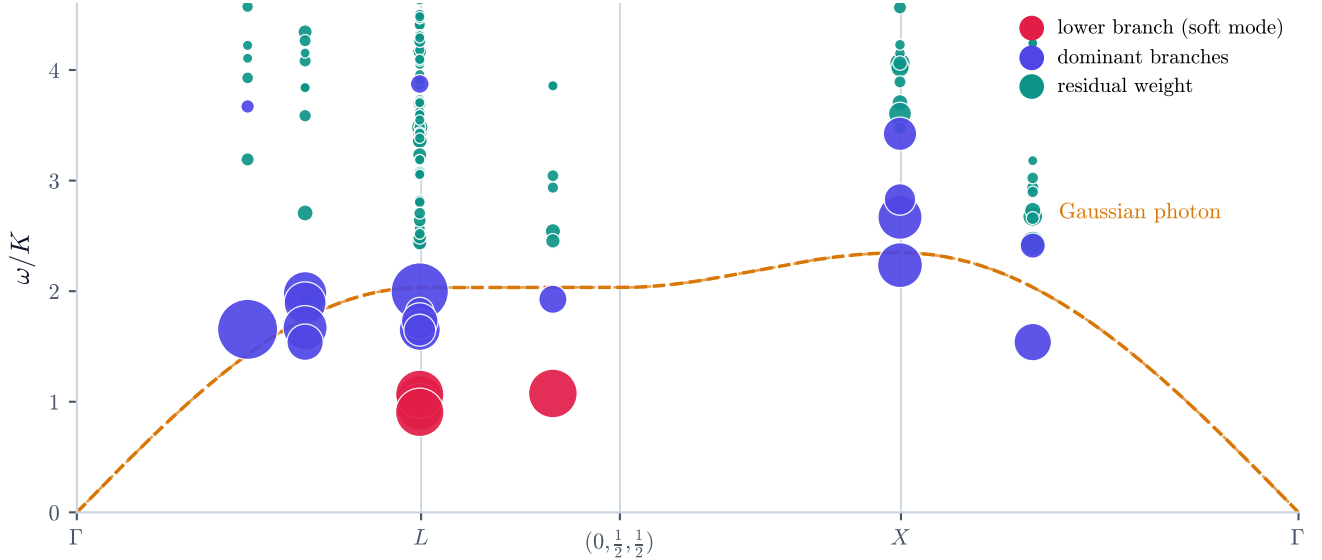


FIG. 2. **The spectrum at lattice momenta.** Operator-resolved S^z spectrum along $\Gamma \rightarrow L \rightarrow (0, \frac{1}{2}, \frac{1}{2}) \rightarrow X \rightarrow \Gamma$, combining all four clusters; marker area is the fraction of the sublattice-summed weight carried by each level. The dashed curve is the Gaussian photon at a single fitted scale, whose two transverse polarizations are exactly degenerate everywhere. Near Γ the two dominant levels track it and each other. Along $\Gamma \rightarrow L$ the lower branch first rises above it and then turns over, falling to half the Gaussian energy at the zone boundary while absorbing half the response.

The parallel stops at the mechanism. In helium Galilean invariance fixes $f_1 = \hbar^2 k^2 / 2m$, so the numerator carries no structure and a roton minimum can only come from $S(k)$ peaking. Here both factors move: against X , S is enhanced by 36%—helium’s mechanism—while f_1 is suppressed by 22% through an exact lattice cancellation

that has no continuum counterpart, and the counterfactual above shows the second is what makes L the minimum. Roughly half of the effect is a mechanism a liquid cannot have. We therefore use “roton-like” descriptively and rest no claim on it.

Experimental implications.—The signature is a two-

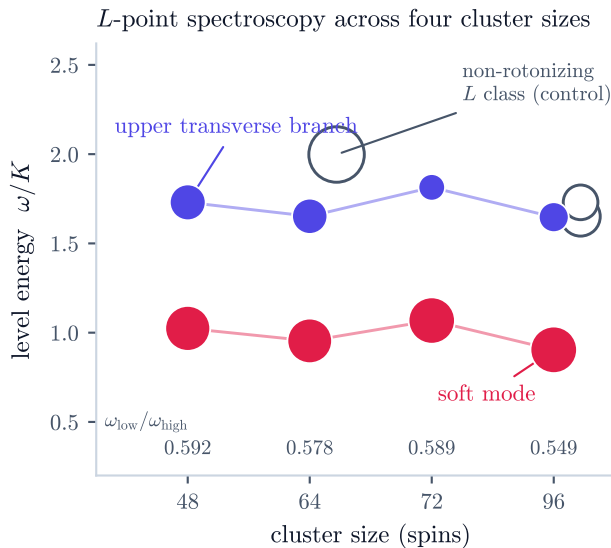


FIG. 3. **Finite-size stability and internal controls.** Filled: the soft level and its partner at L . Open: dominant levels of the inequivalent nonreconstructing class at identical $|\mathbf{q}|$ on the anisotropic 64- and 96-site clusters. Numbers give $\omega_{\text{low}}/\omega_{\text{high}}$.

feature zone-boundary response where a renormalized photon would give one. Following $[111]$ outward, a single dispersing mode at small $\mathbf{q}$ separates into two, the splitting growing from unresolvable to a factor 1.7 at the zone boundary, with the *lower* feature carrying two to three times the weight of the upper. Two aspects are scale-free and so testable without knowing K : the energy ratio, 0.55–0.59, and the weight ratio. A renormalized photon velocity produces neither, and no Gaussian theory produces a splitting at all. Because the effect tracks f_1/S rather than the label L , the same signature should appear along $(t, \frac{1}{2}, \frac{1}{2})$, as it does in our 72-site spectrum.

Two caveats bound this. A material-specific cross section reweights the sublattice amplitudes coherently, so the theoretical 50–54% fraction is not a universal neutron intensity; and longer ring exchanges and virtual spinons can modify the zone-boundary spectrum of a real material. The robust prediction of the minimal model is the redistribution itself.

Discussion.—Harmonic lattice electrodynamics fails at the lattice scale in quantum spin ice, and it fails by order one in three independent, parameter-free respects at once. The reason is that the emergent electric field is saturated. A harmonic theory builds a field wave by raising an amplitude, and the resulting single-mode state is already an eigenstate with nothing left to optimize. A compact theory has no amplitude to raise and must instead choose among exponentially many configurations

carrying the same wave—and it chooses well, re-phasing the ring resonances the wave would dephase and depositing what remains where the probe is blind. The lattice decides where that is possible: at $\mathbf{q}$ parallel to a $\langle 111 \rangle$ axis, one hexagon family in four lies in the wavefronts and drops out of the oscillator strength entirely.

What this is not is a new excitation. It is the emergent photon, reorganized where compactness and geometry together allow it. The open questions are whether the reorganization survives virtual spinons and longer ring exchanges in microscopic XXZ models, and whether the finite-size pole remains sharp in the thermodynamic limit—which requires dynamical rather than ground-state Monte Carlo, since beyond 96 spins we constrain the bound and not the pole.

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- [1] M. Hermele, M. P. A. Fisher, and L. Balents, *Phys. Rev. B* **69**, 064404 (2004).
- [2] C. Castelnovo, R. Moessner, and S. L. Sondhi, *Nature* **451**, 42 (2008).
- [3] M. J. P. Gingras and P. A. McClarty, *Rep. Prog. Phys.* **77**, 056501 (2014).
- [4] L. Savary and L. Balents, *Rep. Prog. Phys.* **80**, 016502 (2017).
- [5] J. G. Rau and M. J. P. Gingras, *Annu. Rev. Condens. Matter Phys.* **10**, 357 (2019).
- [6] C. Broholm, R. J. Cava, S. A. Kivelson, D. G. Nocera, M. R. Norman, and T. Senthil, *Science* **367**, eaay0668 (2020).
- [7] O. Benton, O. Sikora, and N. Shannon, *Phys. Rev. B* **86**, 075154 (2012).
- [8] N. Shannon, O. Sikora, F. Pollmann, K. Penc, and P. Fulde, *Phys. Rev. Lett.* **108**, 067204 (2012).
- [9] Y. Kato and S. Onoda, *Phys. Rev. Lett.* **115**, 077202 (2015).
- [10] K. A. Ross, L. Savary, B. D. Gaulin, and L. Balents, *Phys. Rev. X* **1**, 021002 (2011).
- [11] R. Sibille *et al.*, *Nat. Phys.* **14**, 711 (2018).
- [12] J. Gaudet *et al.*, *Phys. Rev. Lett.* **122**, 187201 (2019).
- [13] B. Gao *et al.*, *Nat. Phys.* **15**, 1052 (2019).
- [14] E. M. Smith *et al.*, *Phys. Rev. X* **12**, 021015 (2022).
- [15] V. Porée, H. Yan, F. Desrochers, S. Petit, E. Lhotel, M. Appel, J. Ollivier, Y. B. Kim, A. H. Nevidomskyy, and R. Sibille, *Nat. Phys.* **21**, 83 (2025).
- [16] B. Gao, F. Desrochers, D. W. Tam, *et al.*, *Nat. Phys.* **21**, 1203 (2025).
- [17] M. P. Kwasigroch, B. Douçot, and C. Castelnovo, *Phys. Rev. B* **95**, 134439 (2017).

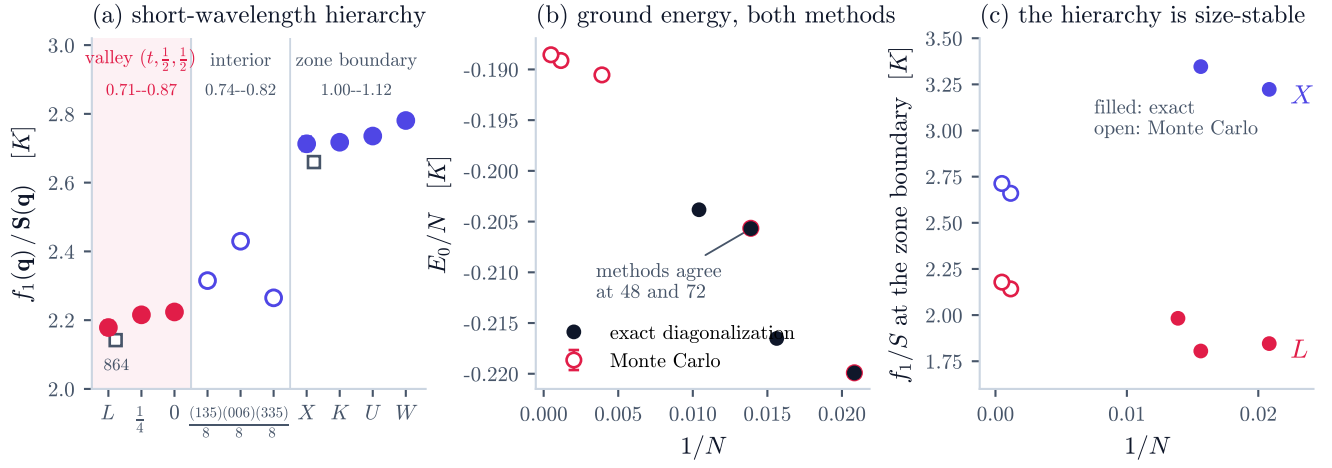


FIG. 4. **The ground-state diagnostic at large size.** (a) $f_1(\mathbf{q})/S(\mathbf{q})$ on the 2048-site supercell grouped by geometric role, with each group's $|\mathbf{q}|$ range indicated; at matched $|\mathbf{q}| = 0.75$ the valley sits 9–10% below the interior star $(0, 0, \frac{3}{4})$. Open squares: 864-site values at shared momenta. (b) Ground-state energy from exact diagonalization and Monte Carlo. (c) Size evolution at L and X across both methods.